Research Manuscript - ICISC 2026

Bias-Free AI-Driven Placement Automation Using Rule-Augmented Machine Learning

---

ICISC 2026 Conference Manuscript Notice
Title: Bias-Free AI-Driven Placement Automation Using Rule-Augmented Machine Learning
Author: Arjun R (Department of Artificial Intelligence, SRM Institute of Science and Technology)
Venue: International Conference on Intelligent Systems and Computing (ICISC 2026)

ABSTRACT:
Automated campus placement screening algorithms frequently inherit demographic selection disparities present in historical recruitment data. This research develops a rule-augmented machine learning framework incorporating Fairlearn mitigation algorithms to bound demographic parity disparities without compromising selection precision across an empirical cohort of 1,200 candidates and 50 recruiting firms.

REPORTED EMPIRICAL METRICS:
- Initial Demographic Parity Gap: 17.9% (Baseline Unmitigated Model)
- Final Demographic Parity Gap: 2.3% (Rule-Augmented Fairlearn Mitigation)
- Reported Hiring Accuracy: 92.8% (Test Cohort Classification Validity)
- Relative Disparity Improvement: 87.2% reduction in demographic selection gap

PUBLICATION STATUS:
This pre-print manuscript placeholder is pending official conference camera-ready proceedings publication.
For full manuscript inquiries or review access, contact: arjunr3773@gmail.com